FloodLens Pakistan: dataset proposal

HackInsight 2026 · team Spark H · Track: climate and heat-health (disaster risk), with a secondary food-security model · Repo: github.com/RoshaanT1/floodlens-pakistan · Dashboard: huggingface.co/spaces/RoshaanT1/floodlens-pakistan

1. Problem, user and what the data must support

District and PDMA planners in Pakistan need 24–72 hours of warning, per district, before dangerous rain, to pre-position teams and supplies. The 2025 monsoon killed more than 1,000 people (NDMA). The data therefore has to give (a) what weather models forecast 1–3 days before each day, exactly as they were issued, and (b) what rain actually fell in each district, independently of those models, over enough years to set each district's own "heavy rain" threshold.

2. Sources used to train and evaluate the models

Source	Publisher and why it is credible	What we use	Coverage / rows in the warehouse	Licence
Open-Meteo Previous Runs API (forecasts issued 1–3 days ahead)	Open-Meteo relays the operational models of ECMWF, NOAA (GFS) and DWD, the national and European weather centres. Previous Runs keeps each run as issued, so features never include later information.	Model A inputs: rain, humidity, pressure, cloud and wind at each district centre, leads 1–3 days	Feb 2024 to Oct 2026; **476,640** district-day-lead rows (`forecast_day`)	CC BY 4.0
NASA GPM IMERG V07 Late Run, daily (GPM_3IMERGDL)	NASA Goddard (GES DISC); the standard multi-satellite rainfall product, used by NDMA-type agencies and the research community; produced independently of the forecast models.	Labels, each district's 95th-percentile threshold (2015–2023), climatology baseline, observed-rain inputs lagged ≥14 h	1 Jan 2015 to 16 Oct 2026, 4,307 days; **689,120** district-days (`rain_district_day`); 163,200 labelled days from 2024	NASA EOSDIS open data policy
OCHA COD-AB admin-2 boundaries (via HDX)	UN OCHA Common Operational Dataset, the boundary set used across the humanitarian system in Pakistan.	160 districts; turning satellite pixels into district values	160 districts	CC BY-IGO
Pakistan Bureau of Statistics weekly SPI, city-wise annex	The official statistics agency's weekly consumer prices.	Model B target and inputs (secondary model)	Jan 2023 to Oct 2026, 169 weeks, 17 cities × 51 items; **146,523** rows (`spi_price`)	Public statistics

Total in the processed layer: over 1.4 million rows from four independent publishers. Model A training rows: 247,680 (to 30 Jun 2025); validation: 88,320 (1 Jul to 31 Dec 2025); test: Jan to Sep 2026, run once.

3. Supporting sources (evaluation groups and citations, not model training targets)

Source	Publisher	Use	Licence
2022 satellite-detected flood extent per district	UNOSAT (UNITAR), via HDX	Flood-prone group (≥10% of the district flooded); model input	CC BY-SA
Elevation (Copernicus DEM GLO-90)	Copernicus via the Open-Meteo Elevation API	Mountain vs plain group; model input	CC BY 4.0
1:10m coastline	Natural Earth	Coastal vs inland group	Public domain
Monsoon 2025 situation reports and advisories (157 documents)	NDMA Pakistan	Insight Agent citations; local validation of flagged districts	NDMA website terms; stored locally, not redistributed

4. Why these sources fit the problem

- **Independence:** inputs come from numerical weather models, labels from satellite rainfall: different publishers and methods, so the evaluation is not circular.
- **No look-ahead:** archived forecasts are stored as issued, and observed rain is used only from days that ended ≥14 h before issue (IMERG Late latency). A test proves this (`tests/test_model_a.py`).
- **Local fit:** thresholds are per district, so "heavy" means unusual for that district, not a national cut-off.

- **Provenance:** every raw file is stored untouched with a manifest row (source, exact query or URL, UTC fetch time, SHA-256, licence); `make data` refuses to build if any hash changed.

5. Splits, baselines and metrics (fixed before results)

	Model A: district heavy-rain risk	Model B: weekly food prices
Target	Daily wettest-pixel rain above the district's 95th percentile of wet-day (≥1 mm) rain, IMERG 2015–2023	Next week's % change in the city average price
Baseline	District-by-month climatology	No change
Metric	Brier score at 24 h (48, 72 h also), 95% bootstrap intervals	MAE of % change; 80% interval coverage
Splits	Train to 30 Jun 2025 · validate 1 Jul – 31 Dec 2025 · test 1 Jan – 30 Sep 2026, run once	

6. Data quality

A Data Quality Agent (pandera) checks schema, ranges, missing values, duplicates and timeliness for every table. Each failed check becomes a **fix, fill or flag** decision, logged with rule, reason, row count and time in `quality_log`, and replayed on rebuild. IMERG fill values (−9999.9) are never read as zero; low sample counts mark a day low-confidence; re-issued days keep the earlier value and are flagged. PBS prices that were not collected stay NULL and are never filled.

7. Known limits of the data

- Daily 0.1° satellite rain smooths out local cloudbursts: on 15 Aug 2025 IMERG's wettest Buner pixel (43 mm) stayed below Buner's threshold (57.5 mm) despite a deadly flash flood.
- Forecasts are taken at district centroids; the label uses the district's wettest pixel.
- The Open-Meteo archive starts in Feb 2024, so Model A trains on 17 months.
- PBS weeks before 30 Apr 2025 come from Internet Archive copies of PBS's old site (provenance recorded per file); a few weeks have no copy.

8. Reserve sources (declared, not used)

ERA5 rainfall via Open-Meteo, only if IMERG access failed (it did not); NOAA GHCN-Daily station data for cross-checks; 2023 census population for reporting exposure.